SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks **COURSE CODE:** CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

(192472188) 7/1/20

I M. Madhulika *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student M. Madhulika

Experiments

QUESTIONS:4 **Total Marks:40**

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

EXPERIMENT-1:

Aim:

To capture and analyze the TCP Three-Way Handshake using Wireshark.

Procedure:

1. Open Wireshark.
2. Select the active Wi-Fi interface and start packet capture.
3. Open a web browser and visit a website (e.g., https://google.com).
4. Stop the packet capture after the webpage loads.
5. Apply the display filter:

tcp.stream == 9

(Your stream number is 9.)

6. Observe the TCP packets containing the **SYN**, **SYN-ACK**, and **ACK** flags.

Observation:

The captured TCP connection contains the following three packets:

- **Packet 711:** SYN – The client (172.25.55.66) initiates the connection by sending a SYN packet to the server (52.123.253.73).
- **Packet 744:** SYN, ACK – The server responds with a SYN-ACK packet, acknowledging the client's request and synchronizing its own sequence number.
- **Packet 746:** ACK – The client sends an ACK packet to acknowledge the server's response, completing the TCP connection establishment.

Role of Each Flag

1. SYN (Synchronize)

- Initiates the TCP connection.
- Synchronizes the client's initial sequence number.
- Sent by the client.

2. SYN-ACK (Synchronize + Acknowledge)

- Sent by the server.
- Acknowledges the client's SYN packet.
- Synchronizes the server's sequence number.

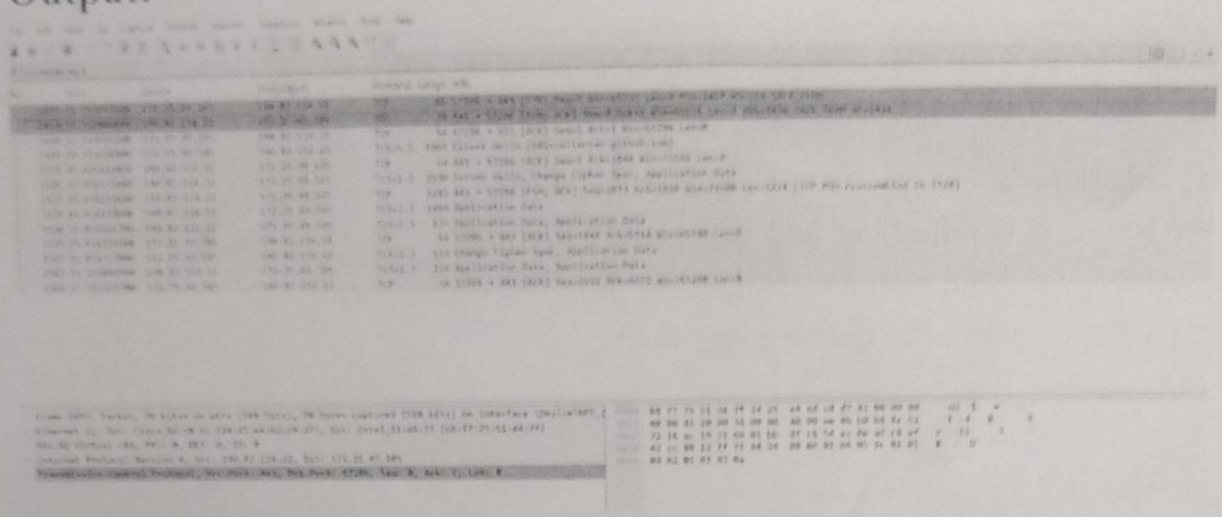
3. ACK (Acknowledge)

- Sent by the client.
- Acknowledges the server's SYN-ACK packet.
- Completes the three-way handshake.

Explanation:

The TCP Three-Way Handshake is used to establish a reliable communication session between a client and a server. First, the client sends a SYN packet requesting a connection. The server replies with a SYN-ACK packet, confirming the request and sending its own synchronization information. Finally, the client responds with an ACK packet, confirming receipt of the server's response. Once these three packets are exchanged, the connection is established, and reliable data transfer can begin.

Output:



Conclusion:

The Wireshark capture successfully demonstrates the TCP Three-Way Handshake using the **SYN**, **SYN-ACK**, and **ACK** packets. This process establishes a reliable connection, synchronizes sequence numbers, and ensures that both the client and server are ready for secure and reliable communication.

EXPERIMENT-2

Aim:

To capture a DNS query using UDP in Wireshark, compare the UDP header size with TCP, and observe the absence of sequence numbers and acknowledgments.

Procedure:

1. Open Wireshark.
2. Select the active Wi-Fi interface.
3. Start packet capture.
4. Open Command Prompt and execute:

nslookup google.com

5. Stop the packet capture.
6. Apply the display filter:

udp.port == 53

7. Select the DNS query or response packet.
8. Expand the **User Datagram Protocol (UDP)** and **Domain Name System (DNS)** sections to view the packet details.

Observation:

The captured packet shows that DNS communication uses the **UDP protocol** over port 53. The UDP header contains only the following fields:

- Source Port
- Destination Port
- Length
- Checksum

Unlike TCP, the UDP header **does not contain Sequence Number or Acknowledgment Number fields**, making it much simpler and smaller.

UDP vs TCP Header Comparison

Feature	UDP	TCP
Header Size	8 bytes	20 bytes (minimum)
Source Port	Yes	Yes
Destination Port	Yes	Yes
Sequence Number	No	Yes
Acknowledgment Number	No	Yes
Checksum	Yes	Yes
Reliability	No	Yes
Connection Type	Connectionless	Connection-oriented
Speed	Faster	Slower

Explanation:

The Wireshark capture confirms that DNS uses **UDP** because DNS messages are generally small and require fast transmission. UDP has a fixed **8-byte header**, while TCP has a minimum **20-byte header**. Since UDP does not establish a connection or maintain sequence numbers and acknowledgments, it reduces communication overhead and allows DNS queries to be completed quickly. However, because UDP does not retransmit lost packets or guarantee delivery, it is less reliable than TCP.

Impact on Speed and Reliability

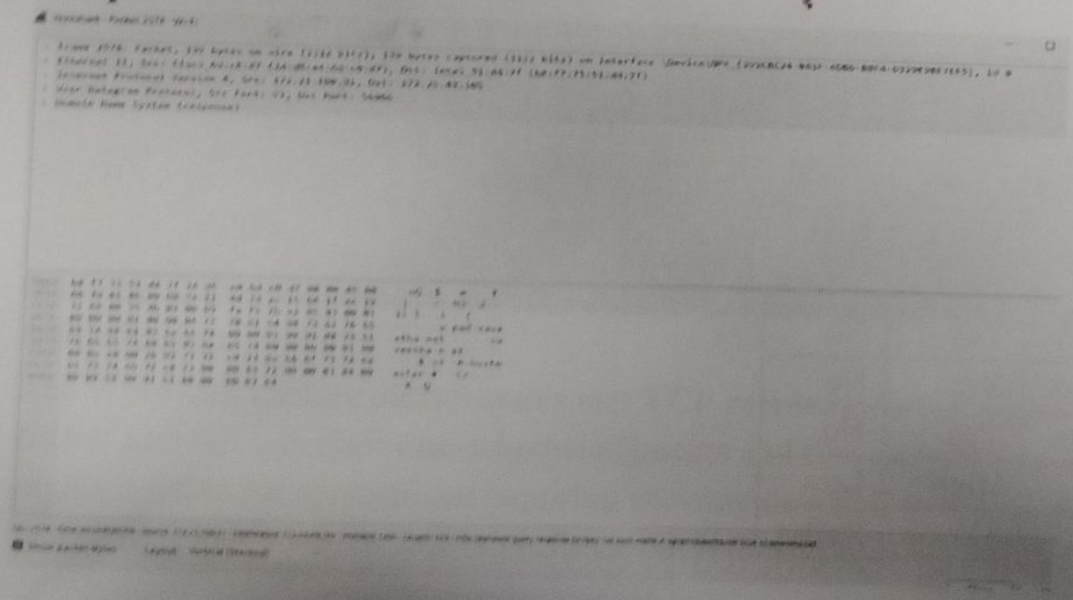
Speed

- Small **8-byte header**.
- No connection establishment.
- No acknowledgments.
- Low protocol overhead.
- Faster transmission and lower latency.

Reliability

- No sequence numbers.
- No acknowledgments.
- No retransmission of lost packets.
- Packets may be lost or arrive out of order.
- Less reliable than TCP.

Output:



Conclusion:

The experiment demonstrates that **DNS typically uses UDP** because it is lightweight and efficient. The absence of sequence numbers and acknowledgments, along with its small **8-byte header**, allows faster communication. However, this design sacrifices reliability compared to TCP, making UDP suitable for applications where speed is more important than guaranteed delivery.

EXPERIMENT-3

Aim:

To simulate packet loss, observe TCP retransmission in Wireshark, and compare TCP with UDP in terms of reliability and speed.

Procedure:

1. Open **Wireshark**.
2. Select the active **Wi-Fi** interface and start packet capture.
3. Open a website or start downloading a file.
4. Briefly disconnect the Wi-Fi connection for **5–10 seconds**.
5. Reconnect the Wi-Fi.
6. Stop the packet capture.
7. Apply the display filter:

tcp

8. Observe packets marked as:
   - **TCP Retransmission**
   - **TCP Spurious Retransmission**
   - **TCP Dup ACK**
9. Apply the filter:

udp

10. Observe that UDP packets are **not retransmitted**.

Observation:

The Wireshark capture shows several **TCP Retransmission**, **TCP Spurious Retransmission**, and **TCP Duplicate ACK** packets after the temporary network interruption. These packets indicate that some data packets were lost during transmission. Since the sender did not receive the expected acknowledgment, TCP retransmitted the missing packets. When observing UDP traffic, no retransmission packets were found because UDP does not provide a mechanism for detecting or recovering lost packets.

TCP vs UDP During Packet Loss

Feature	TCP	UDP
Detects Packet Loss	Yes	No
Retransmits Lost Packets	Yes	No
Uses Sequence Numbers	Yes	No
Uses Acknowledgments	Yes	No
Reliability	High	Low
Speed	Slower	Faster

Explanation:

TCP is a connection-oriented protocol that ensures reliable communication. It uses **sequence numbers** to track transmitted packets and **acknowledgments (ACKs)** to confirm successful delivery. If an acknowledgment is not received within a certain time, TCP assumes that the packet has been lost and retransmits it. This guarantees reliable delivery but increases communication delay and network overhead.

UDP is a connectionless protocol that does not use acknowledgments, sequence numbers, or retransmissions. If a UDP packet is lost, it is simply discarded without being resent. This reduces overhead and provides faster communication but does not guarantee reliable delivery.

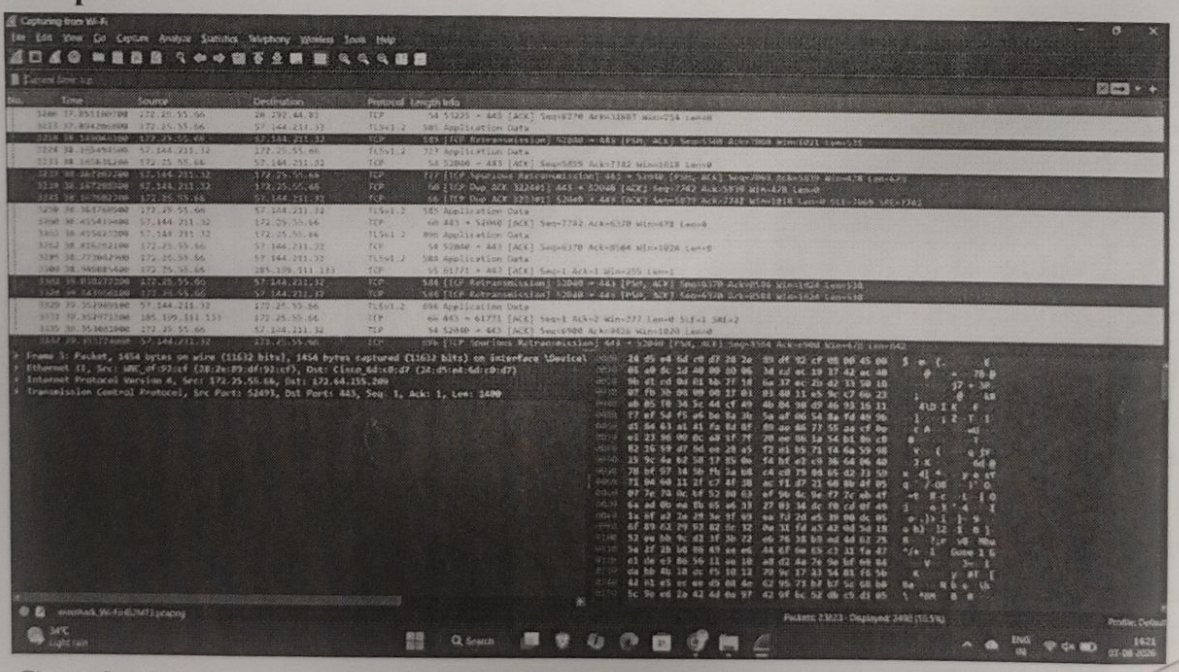
Why TCP is Reliable but Slower

- Uses acknowledgments to confirm delivery.
- Detects lost packets.
- Retransmits missing packets.
- Maintains packet order using sequence numbers.
- Introduces extra delay due to retransmissions and error recovery.

Why UDP is Faster but Less Reliable

- No connection setup.
- No acknowledgments.
- No retransmissions.
- Smaller protocol overhead.
- Lost packets are not recovered.

Output:



Conclusion:

The Wireshark capture demonstrates that **TCP retransmits lost packets** after detecting packet loss through duplicate acknowledgments and timeout mechanisms. This makes TCP highly reliable for applications requiring accurate data delivery. In contrast, **UDP does not retransmit lost packets**, making it faster but less reliable. Therefore, TCP is preferred for

applications such as web browsing and file transfer, while UDP is suitable for real-time applications such as video streaming, online gaming, and voice communication.

EXPERIMENT-4:

Aim:

To capture DNS traffic using Wireshark while resolving a domain name, identify the DNS query type and response, measure the DNS resolution time, and explain why DNS typically uses UDP.

Procedure:

1. Open **Wireshark**.
2. Select the active **Wi-Fi** interface.
3. Start packet capture.
4. Open Command Prompt and execute:

nslookup google.com

5. Stop the packet capture.
6. Apply the display filter:

dns

7. Observe the DNS query and DNS response packets.
8. Identify the query type (**A**, **AAAA**, or **MX**) and the returned response.
9. Measure the time taken between the query and the response.

Observation:

The Wireshark capture shows DNS queries and their corresponding responses.

Query Type A

Field	Value
Domain Name	google.com
Query Type	A (IPv4 Address)
Response	**142.250.182.78**

Query Type AAAA

Field	Value
Domain Name	google.com
Query Type	AAAA (IPv6 Address)
Response	**2404:6800:4007:835::200e**

DNS Resolution Time

From your capture:

- A Query Time: 14.180772300 s
- A Response Time: 14.486452600 s

Resolution Time = Response Time − Query Time

= 14.486452600 − 14.180772300

= 0.3056803 seconds

≈ 306 ms

Why DNS Typically Uses UDP

DNS generally uses **UDP** because:

- It is **connectionless**, so there is no connection setup.

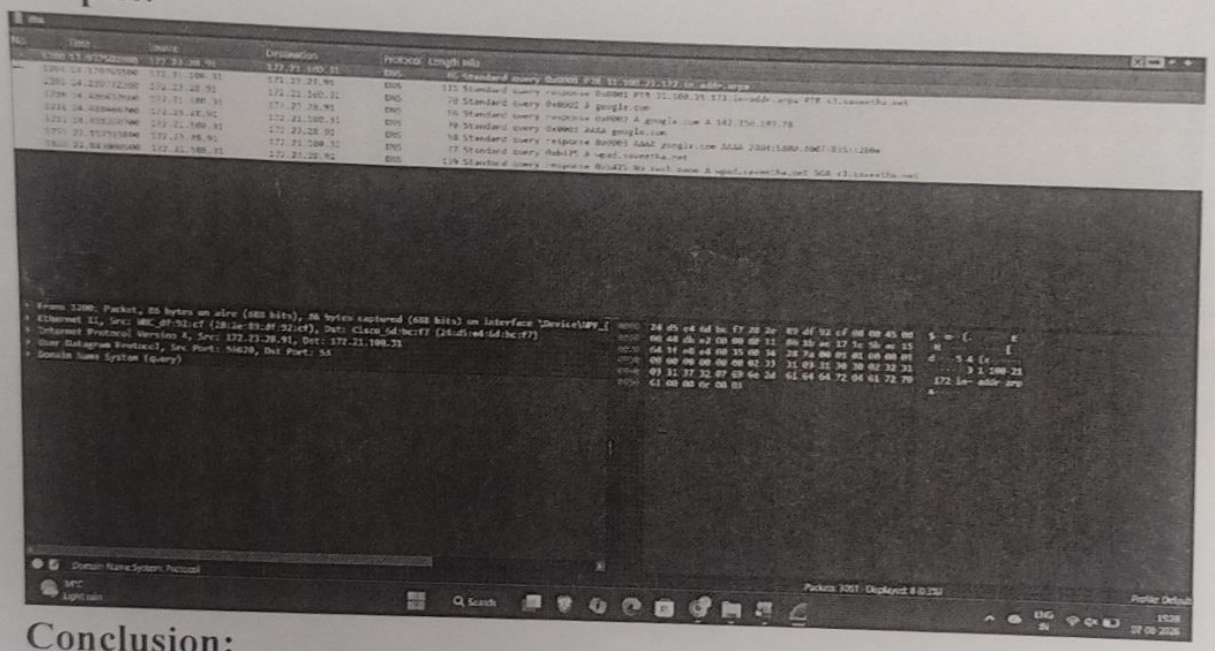
- It has a small **8-byte header**, reducing overhead.
- DNS requests and responses are usually small.
- It provides **fast communication** with low latency.
- It reduces network traffic and improves efficiency.

When DNS responses are too large or for operations such as zone transfers, **TCP** is used instead.

Explanation:

The Wireshark capture shows that the client sends a DNS query to the DNS server requesting the IP address of **google.com**. The DNS server responds with the requested record. In this capture, both an **A record** (IPv4 address) and an **AAAA record** (IPv6 address) are returned. The time difference between the query and the response represents the DNS resolution time. DNS uses UDP because it is lightweight, connectionless, and provides fast name resolution with minimal overhead.

Output:



Conclusion:

The experiment successfully captured DNS traffic while resolving **google.com**. The DNS server returned an **A record (142.250.182.78)** and an **AAAA record (2404:6800:4007:835::200e)**. The measured DNS resolution time was approximately **306 ms**. DNS typically uses **UDP** because it offers faster communication with lower overhead, making it suitable for quick domain name resolution.